

Resilient Cooperative Guidance Under Hybrid DoS and False-Data Injection Attacks

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Abstract—This draft presents a unified treatment for denial-of-service (DoS) and false-data injection (FDI) attacks in networked cooperative guidance of multiple flight vehicles. DoS attacks remove communication links and make neighbor information unavailable, whereas FDI attacks keep the communication channel active but corrupt the transmitted guidance variables. To handle the two attack modes within a single framework, a trusted-graph isolation mechanism is proposed. The mechanism first detects link availability, then evaluates data credibility through a multi-evidence trust mechanism that combines physical feasibility, group-consistency clustering, and windowed distribution divergence. Once a link is judged to be corrupted by FDI, it is removed from the cooperative guidance graph rather than softly blended with a substitute value. A resilient impact-time synchronization error is then defined on the remaining trusted graph. Exponential and fixed-time convergence conditions are derived in terms of accumulated trusted connectivity after DoS blocking and FDI isolation.

Index Terms—Cooperative guidance, denial-of-service attack, false-data injection attack, resilient control, virtual leader, finite-time convergence.

I. INTRODUCTION

NETWORKED cooperative guidance enables multiple flight vehicles to coordinate their impact times by exchanging local guidance variables, such as time-to-go estimates or impact-time errors. In an ideal communication environment, this problem can be converted into a consensus problem over a connected graph. However, the communication graph may be damaged by cyber attacks. In particular, DoS attacks interrupt communication links and make neighbor information unavailable, whereas FDI attacks inject corrupted data into still-active communication links. These two attack modes create different difficulties: DoS causes missing information, while FDI causes misleading information.

A direct zero-bias strategy can be used during DoS intervals. Under such a strategy, the cooperative terms associated with disconnected links are removed, and the guidance law degenerates to a local or proportional-navigation-like behavior. This approach is conservative and useful for stability preservation, but it suspends part of the cooperation and may introduce discontinuities in the command. Conversely, a single-predictor residual detector may attenuate simple FDI signals but can be unreliable in cooperative guidance because target maneuvers, modeling errors, field-of-view constraints, and intermittent DoS intervals can all enlarge the prediction residual.

This draft therefore proposes a unified trusted-graph framework. The central idea is to decide which neighbor information is allowed to enter the cooperative guidance law. DoS is

handled through link-availability indicators, and detected FDI is handled by removing the corrupted link from the cooperative graph. A zero-bias fallback is used when the remaining trusted graph is not connected, so unavailable or corrupted links cannot inject positive energy into the impact-time disagreement dynamics.

The one-sentence argument of this draft is as follows: in cooperative guidance under hybrid DoS and FDI attacks, the impact-time synchronization error can be driven to zero before interception by isolating unavailable or corrupted links and by requiring sufficient accumulated trusted connectivity before impact.

II. COOPERATIVE GUIDANCE VARIABLE

Consider N flight vehicles intercepting a common target. The communication topology is described by

$$\mathcal{G}(t) = \{\mathcal{V}, \mathcal{E}(t), A(t)\}, \quad (1)$$

where $\mathcal{V} = \{1, 2, \dots, N\}$ is the vehicle set, $\mathcal{E}(t)$ is the communication edge set, and $A(t) = [a_{ij}(t)]$ is the adjacency matrix. If vehicle i can receive information from vehicle j , then $a_{ij}(t) > 0$; otherwise $a_{ij}(t) = 0$.

The main cooperative variable is selected as the time-to-go estimate. Let

$$t_{go,i}(t) = t_{f,i}(t) - t \quad (2)$$

denote the remaining flight time of vehicle i , where $t_{f,i}$ is its predicted impact time. Simultaneous interception requires

$$t_{go,1}(t) = t_{go,2}(t) = \dots = t_{go,N}(t). \quad (3)$$

Without attacks, the local cooperative error can be defined as

$$e_i(t) = \sum_{j \in \mathcal{N}_i(t)} a_{ij}(t) [t_{go,i}(t) - t_{go,j}(t)], \quad (4)$$

where $\mathcal{N}_i(t)$ is the neighbor set of vehicle i .

III. HYBRID DoS AND FDI ATTACK MODEL

Let $y_{ij}(t)$ denote the information received by vehicle i from vehicle j . Under hybrid DoS and FDI attacks, the received signal is modeled as

$$y_{ij}(t) = \gamma_{ij}(t) [x_j(t) + f_{ij}(t)], \quad (5)$$

where $x_j(t)$ is the true transmitted information of vehicle j , $f_{ij}(t)$ is the FDI signal injected into link (j, i) , and $\gamma_{ij}(t)$ is the DoS switching signal:

$$\gamma_{ij}(t) = \begin{cases} 1, & \text{link } (j, i) \text{ is available,} \\ 0, & \text{link } (j, i) \text{ is blocked by DoS.} \end{cases} \quad (6)$$

For time-to-go coordination, (5) becomes

$$y_{ij}(t) = \gamma_{ij}(t) [t_{go,j}(t) + f_{ij}(t)]. \quad (7)$$

This model covers three cases. If $\gamma_{ij} = 1$ and $f_{ij} = 0$, the link is normal. If $\gamma_{ij} = 1$ and $f_{ij} \neq 0$, the link is active but corrupted by FDI. If $\gamma_{ij} = 0$, the link is unavailable due to DoS.

IV. FDI TREATMENT: MULTI-EVIDENCE TRUST EVALUATION

FDI attacks should not be handled as missing data because the corrupted signal is still received. A direct approach is to compare the received information with a local prediction. Let $\hat{t}_{go,j}^i(t)$ denote the prediction of vehicle j 's time-to-go maintained by vehicle i . The residual is

$$r_{ij}(t) = y_{ij}(t) - \hat{t}_{go,j}^i(t). \quad (8)$$

However, a detector based only on $r_{ij}(t)$ is fragile in cooperative guidance. Target maneuvers, modeling errors, field-of-view constraints, previous DoS intervals, and inaccurate time-to-go approximation can all enlarge the residual even when no attack exists. Conversely, an intelligent attacker may inject data that remain close to the predictor and therefore bypass a single residual threshold. For this reason, the residual is treated only as one weak evidence source, and the credibility of each link is evaluated by a multi-evidence trust mechanism.

The proposed FDI treatment answers three questions for each received neighbor datum:

- Is the datum physically feasible under the engagement and guidance constraints?
- Is the datum consistent with the information reported by other neighbors?
- Does the recent residual distribution deviate from the nominal residual distribution?

The three answers are converted into continuous trust scores and fused into the final link trust weight $\rho_{ij}(t) \in [0, 1]$.

A. Physical Feasibility Evidence

The first evidence source uses guidance-domain constraints instead of a precise predictor. Based on the last trusted state of vehicle j , the maximum speed or acceleration, the admissible field-of-view region, and the remaining range, vehicle i constructs a feasible interval for the neighbor's time-to-go:

$$t_{go,j}(t) \in [\underline{t}_{go,j}^i(t), \bar{t}_{go,j}^i(t)]. \quad (9)$$

If the received value lies outside this interval, it violates basic engagement feasibility and is strongly suspicious. The physical trust score can be defined as

$$\rho_{ij}^{\text{phy}}(t) = \begin{cases} 1, & y_{ij}(t) \in [\underline{t}_{go,j}^i(t), \bar{t}_{go,j}^i(t)] \\ \epsilon_\rho, & \text{otherwise,} \end{cases} \quad (10)$$

where $0 < \epsilon_\rho \ll 1$ avoids an unnecessarily hard rejection. This layer is robust to moderate prediction errors because it only checks whether the received datum is physically possible.

B. Group-Consistency or GMM Evidence

The second evidence source uses the redundancy of the multi-vehicle system. Under normal cooperation, the information reported by neighboring vehicles should be mutually consistent. Define a feature vector

$$\xi_j(t) = [t_{go,j}(t) \quad \dot{t}_{go,j}(t) \quad q_j(t) \quad \dot{q}_j(t)]^T, \quad (11)$$

where q_j and $\dot{q}_j$ denote line-of-sight-related variables. The neighbor feature set observed by vehicle i is

$$\mathcal{X}_i(t) = \{\xi_\ell(t) : \ell \in \mathcal{N}_i(t)\}. \quad (12)$$

A Gaussian mixture model can cluster $\mathcal{X}_i(t)$ into a normal component and a suspicious component. The corresponding trust score is the belief that vehicle j belongs to the normal component:

$$\rho_{ij}^{\text{gmm}}(t) = \Pr(\xi_j(t) \in \mathcal{C}_{\text{normal}} \mid \mathcal{X}_i(t)). \quad (13)$$

This idea follows the logic of clustering-based secure state estimation: rather than assuming a known attack form, the detector lets the data separate normal and abnormal information.

If a full GMM is too costly or the number of neighbors is small, a median-consistency score can be used instead:

$$d_{ij}^{\text{med}}(t) = |y_{ij}(t) - \text{median}_{\ell \in \mathcal{N}_i(t)} y_{i\ell}(t)|, \quad (14)$$

$$\rho_{ij}^{\text{grp}}(t) = \exp[-\kappa_g d_{ij}^{\text{med}}(t)], \quad \kappa_g > 0. \quad (15)$$

The larger the deviation from the group-consistent value is, the lower the trust becomes.

C. Windowed Distribution Evidence

The third evidence source evaluates the recent statistical behavior of the residual. This is important because FDI attacks may be slow, intermittent, or small enough to avoid a single-time threshold. Define a residual window

$$\mathcal{R}_{ij}^L(t) = \{r_{ij}(t-L+1), \dots, r_{ij}(t)\}, \quad (16)$$

where L is the window length. Let $\mathcal{P}_{ij}^L(t)$ be the empirical distribution of this window and let $\mathcal{P}_{ij}^0$ be the nominal residual distribution obtained from attack-free data or from a recently trusted window. A Kullback–Leibler divergence score is

$$D_{ij}^{\text{KL}}(t) = D_{\text{KL}}(\mathcal{P}_{ij}^L(t) \parallel \mathcal{P}_{ij}^0). \quad (17)$$

The distribution-based trust score is then

$$\rho_{ij}^{\text{kl}}(t) = \exp[-\kappa_{kl} D_{ij}^{\text{KL}}(t)], \quad \kappa_{kl} > 0. \quad (18)$$

A short window reacts quickly but may create more false alarms, whereas a long window is smoother but slower. Thus, L should be selected according to the expected attack duration and guidance update rate.

D. Trust Fusion and Smoothing

The final raw trust score is obtained by fusing the three evidence channels:

$$\bar{\rho}_{ij}(t) = \text{sat}_{[0,1]} \left(\rho_{ij}^{\text{phy}}(t) \rho_{ij}^{\text{gmm}}(t) \rho_{ij}^{\text{kl}}(t) \right). \quad (19)$$

If the median-consistency score is used instead of GMM, ρ_{ij}^{gmm} in (19) is replaced by ρ_{ij}^{grp} .

To avoid abrupt command changes caused by sudden trust switching, the implemented trust weight is filtered as

$$\dot{\rho}_{ij}(t) = -a_\rho [\rho_{ij}(t) - \bar{\rho}_{ij}(t)], \quad a_\rho > 0. \quad (20)$$

The resulting $\rho_{ij}(t)$ is a continuous credibility weight. A value close to one means that the received information is physically feasible, group-consistent, and distributionally normal. A value close to zero means that the link is likely corrupted by FDI. In the control layer, this continuous score is converted into the binary isolation variable $\sigma_{ij}(t)$ used in (22).

Remark 1. During a DoS interval, no new neighbor data are available. Therefore, FDI evidence is frozen or propagated from the last trusted value, and the corresponding link is removed from the trusted graph through $\gamma_{ij} = 0$. This separation avoids confusing missing data with corrupted data.

V. DOS TREATMENT: LINK REMOVAL AND ZERO-BIAS FALLBACK

DoS attacks make neighbor information unavailable. In the proposed trusted-graph formulation, a DoS-blocked link is removed from the cooperative graph by setting $\gamma_{ij}(t) = 0$. The corresponding adjacency entry $a_{ij}^r(t)$ is then zero, so the unavailable datum cannot enter the cooperative guidance command.

This link-removal rule is conservative but important for stability. If no trusted neighbor is available for vehicle i , the cooperative term is set to zero and the vehicle follows the baseline guidance command. In impact-time coordinates, this means that no unverified neighbor reference is injected into the disagreement dynamics during the DoS interval.

For implementation smoothing, a virtual-leader reference $t_{go,v,ij}(t)$ may still be propagated from the last trusted value:

$$t_{go,v,ij}(t_d) = y_{ij}(t_d^-), \quad \dot{t}_{go,v,ij}(t) = -1. \quad (21)$$

However, this reference is allowed to affect the cooperative command only if it satisfies a nonexpansive safety condition. Otherwise, the disconnected edge remains removed. The zero-bias fallback is therefore the default stability-preserving action under DoS.

VI. TRUSTED GRAPH CONSTRUCTION BY LINK ISOLATION

The detection layer is used only to decide whether a received datum is allowed to enter the cooperative guidance law. Once an FDI attack is detected on link (j, i) , the link is removed from the cooperative graph instead of being softly blended with a substitute value. Define the binary isolation variable

$$\sigma_{ij}(t) = \begin{cases} 1, & \text{link } (j, i) \text{ is judged trustworthy,} \\ 0, & \text{FDI is detected on link } (j, i). \end{cases} \quad (22)$$

For the convergence proof, the trusted graph is taken as an undirected graph. This is consistent with bidirectional cooperative communication, or with a symmetrized implementation in which a pairwise cooperative edge is used only when both directional messages are available and trusted. Thus the trusted adjacency entry used in the analysis is

$$a_{ij}^r(t) = a_{ij} \gamma_{ij}^r(t) \sigma_{ij}^r(t), \quad a_{ji}^r(t) = a_{ji} \gamma_{ji}^r(t) \sigma_{ji}^r(t), \quad (23)$$

where $\gamma_{ij}^r(t) = \gamma_{ij}(t) \gamma_{ji}(t)$ and $\sigma_{ij}^r(t) = \sigma_{ij}(t) \sigma_{ji}(t)$ for $i \neq j$. If only directed communication is available, the same controller can still be implemented, but the proof must be replaced by a weight-balanced or strongly connected directed-graph argument. The resulting trusted graph is denoted by $\mathcal{G}_r(t) = (\mathcal{V}, \mathcal{E}_r(t))$. Thus, DoS and detected FDI have the same effect at the control layer: the corresponding edge is excluded. If several outgoing links of the same sender are detected as corrupted, a conservative node-isolation rule may also be used:

$$a_{ij}^r(t) = a_{ij} \gamma_{ij}^r(t) \sigma_i(t) \sigma_j(t), \quad (24)$$

where $\sigma_j(t) = 0$ removes node j from the cooperative graph. The main analysis below is stated for link isolation; node isolation follows by applying the same proof to the remaining normal-node subgraph.

Let

$$T_i(t) = t + t_{go,i}(t) \quad (25)$$

be the predicted impact time of vehicle i . The resilient synchronization error is defined on the trusted graph as

$$e_i^r(t) = \sum_{j \in \mathcal{N}_i} a_{ij}^r(t) [T_i(t) - T_j(t)]. \quad (26)$$

Equivalently, with $T = [T_1, \dots, T_N]^\top$,

$$e^r(t) = L_r(t) T(t), \quad (27)$$

where $L_r(t)$ is the Laplacian matrix of $\mathcal{G}_r(t)$. In this form, corrupted data do not appear in the guidance law after detection.

Remark 2. The virtual-leader variable can still be used for trajectory smoothing or for local fallback during long DoS intervals, but it is not required for excluding FDI data from the stability proof. The rigorous convergence statement below relies on edge removal and trusted-graph connectivity rather than on a soft reconstruction of corrupted neighbor information.

VII. RESILIENT COOPERATIVE GUIDANCE LAW

The cooperative guidance objective is to regulate the predicted impact times T_i . To connect the graph-level controller with the planar guidance command, write the local impact-time dynamics in an input-affine form

$$\dot{T}_i = \phi_i(t) + g_i(t) a_i^{\text{coop}} + d_i(t), \quad (28)$$

where a_i^{coop} is the scalar cooperative normal-acceleration component, $\phi_i(t)$ is the drift induced by the baseline guidance and engagement geometry, $g_i(t)$ is the local impact-time sensitivity to the cooperative acceleration, and $d_i(t)$ collects modeling

error. This representation is obtained by differentiating $T_i = t + t_{go,i}$ along the relative-motion equations and separating the part driven by the cooperative acceleration. The controller is applied only on intervals where

$$0 < \underline{g} \leq |g_i(t)| \leq \bar{g} < \infty. \quad (29)$$

The desired synchronization input is denoted by v_i . The normal-acceleration command can then be selected as

$$a_i^{\text{coop}} = \frac{v_i - \phi_i(t)}{g_i(t)} \quad (30)$$

when $g_i(t)$ is known, or by a robust approximation when $g_i(t)$ is estimated. In the nominal analysis $d_i = 0$, so (28) reduces to $\dot{T}_i = v_i$.

At the abstract synchronization level, the trusted-graph feedback is

$$\dot{T} = -kL_r(t)T, \quad k > 0, \quad (31)$$

for an exponentially convergent design. A fixed-time alternative is

$$\dot{T}_i = -k_1 \sum_{j \in \mathcal{N}_i} a_{ij}^r \text{sig}^\alpha(T_i - T_j) - k_2 \sum_{j \in \mathcal{N}_i} a_{ij}^r \text{sig}^\beta(T_i - T_j), \quad (32)$$

where $k_1, k_2 > 0$, $0 < \alpha < 1$, $\beta > 1$, and $\text{sig}^\mu(x) = |x|^\mu \text{sgn}(x)$. The actual normal-acceleration command is then selected to realize the desired change of T_i through the guidance geometry:

$$\mathbf{a}_i = \mathbf{a}_i^{\text{PNG}} + \mathbf{a}_i^{\text{coop}}, \quad (33)$$

where $\mathbf{a}_i^{\text{PNG}}$ is the baseline interception term and $\mathbf{a}_i^{\text{coop}}$ implements either (31) or (32) through the local relation between normal acceleration and $\dot{T}_i$.

Assumption 1 (Impact-time control channel). *For each non-isolated normal vehicle, the sensitivity $g_i(t)$ satisfies (29) on the time interval considered. In the nominal theorem the compensated channel satisfies $\dot{T}_i = v_i$. Under modeling uncertainty, the compensated channel satisfies $\dot{T}_i = v_i + d_i(t)$ with $\|d(t)\| \leq \bar{d}$.*

VIII. CONVERGENCE UNDER DOS AND DETECTED FDI

The convergence analysis is now a switching-graph problem. DoS removes unavailable links through $\gamma_{ij} = 0$, and detected FDI removes corrupted links through $\sigma_{ij} = 0$. Hence, the only graph that enters the controller is $\mathcal{G}_r(t)$.

Assumption 2 (Trusted connectivity after isolation). *The trusted graph $\mathcal{G}_r(t)$ is undirected and piecewise constant, with finitely many switches on every finite time interval. There exists a constant $\underline{\lambda} > 0$ such that, during trusted connected intervals,*

$$\lambda_2(L_r(t)) \geq \underline{\lambda}. \quad (34)$$

When $\mathcal{G}_r(t)$ is disconnected, $\lambda_2(L_r(t))$ is set to zero in the convergence bound.

Assumption 3 (Nonexpansive fallback). *When the trusted graph is disconnected because of DoS or FDI isolation, the cooperative term is either computed on the remaining trusted connected components or set to zero on unsafe links.*

This fallback does not increase the impact-time disagreement energy.

Define the projection matrix

$$M = I - \frac{1}{N} \mathbf{1}\mathbf{1}^\top \quad (35)$$

and the disagreement vector

$$\tilde{T} = MT. \quad (36)$$

The Lyapunov function is

$$V(t) = \frac{1}{2} \tilde{T}^\top(t) \tilde{T}(t). \quad (37)$$

Theorem 1 (Exponential convergence under trusted connectivity). *Consider the nominal compensated impact-time dynamics $\dot{T} = -kL_r(t)T$ generated by (28)–(30). Suppose that detected FDI links and DoS-blocked links are excluded according to (23). Then the disagreement energy satisfies*

$$V(t) \leq V(0) \exp \left[-2k \int_0^t \lambda_2(L_r(s)) ds \right]. \quad (38)$$

Consequently, if

$$\int_0^\infty \lambda_2(L_r(s)) ds = \infty, \quad (39)$$

then $T_i(t) - T_j(t) \rightarrow 0$ for all non-isolated normal vehicles that belong to the trusted graph. For a prescribed terminal time t_f , the sufficient condition

$$\int_0^{t_f} \lambda_2(L_r(s)) ds \geq \frac{1}{2k} \ln \frac{V(0)}{\varepsilon} \quad (40)$$

guarantees $V(t_f) \leq \varepsilon$.

Proof. Because the trusted graph is undirected, $L_r(t)$ is symmetric positive semidefinite and satisfies $L_r(t)\mathbf{1} = 0$. Hence $ML_r(t) = L_r(t)M = L_r(t)$. During intervals governed by (31),

$$\dot{\tilde{T}} = M\dot{T} = -kML_rT = -kL_r\tilde{T}. \quad (41)$$

Therefore,

$$\dot{V} = \tilde{T}^\top \dot{\tilde{T}} = -k\tilde{T}^\top L_r \tilde{T}. \quad (42)$$

For a connected trusted graph,

$$\tilde{T}^\top L_r \tilde{T} \geq \lambda_2(L_r) \|\tilde{T}\|^2 = 2\lambda_2(L_r)V. \quad (43)$$

Hence,

$$\dot{V} \leq -2k\lambda_2(L_r)V. \quad (44)$$

When the trusted graph is disconnected, $L_r(t)$ is still positive semidefinite, so the same calculation gives $\dot{V} \leq 0$ and $\lambda_2(L_r) = 0$. The nonexpansive fallback therefore preserves this inequality even when unsafe links are removed or the cooperative term is set to zero. Since the switching signal is piecewise constant and has no Zeno behavior, the comparison lemma can be applied on successive intervals and concatenated over time. This gives (38). Conditions (39) and (40) follow directly from (38). $\square$

Remark 3 (Effect of bounded channel error). *If the compensated channel has a bounded residual disturbance, $\tilde{T} = -kL_r T + d(t)$, then*

$$\dot{V} \leq -2k\lambda_2(L_r)V + \|\tilde{T}\| \|Md\|. \quad (45)$$

On intervals where $\lambda_2(L_r) \geq \underline{\lambda}$, Young's inequality gives

$$\dot{V} \leq -k\underline{\lambda}V + \frac{\|Md\|^2}{2k\underline{\lambda}}. \quad (46)$$

Thus bounded impact-time channel error leads to practical synchronization, with an ultimate disagreement bound proportional to $\bar{d}/(k\underline{\lambda})$ rather than exact convergence.

Corollary 1 (Fixed-time convergence with accumulated trusted time). *Under the fixed-time controller (32), suppose that the trusted connected graph satisfies (34) during connected intervals and that the fallback is nonexpansive otherwise. If the controller and the impact-time channel yield constants $c_1, c_2 > 0$ such that*

$$\dot{V} \leq -\delta_r(t) \left[c_1 V^{\frac{1+\alpha}{2}} + c_2 V^{\frac{1+\beta}{2}} \right], \quad (47)$$

where $\delta_r(t) = 1$ on trusted connected intervals and $\delta_r(t) = 0$ otherwise, then V reaches zero once

$$\int_0^t \delta_r(s) ds \geq \frac{2}{c_1(1-\alpha)} + \frac{2}{c_2(\beta-1)}. \quad (48)$$

Proof. For $V > 0$, set $z = \sqrt{V}$. From (47),

$$\dot{z} \leq -\frac{\delta_r(t)}{2} (c_1 z^\alpha + c_2 z^\beta). \quad (49)$$

Using the accumulated trusted time $\tau_r(t) = \int_0^t \delta_r(s) ds$ as the time variable gives

$$\frac{dz}{d\tau_r} \leq -\frac{1}{2} (c_1 z^\alpha + c_2 z^\beta). \quad (50)$$

The standard fixed-time bound yields (48). $\square$

IX. CONNECTION WITH PROBABILISTIC SAFE COMMUNICATION TIME

If the DoS process is random, the safe communication duration can be described probabilistically. For example,

$$T_s = T \sum_{k=0}^{\mathcal{E}} F(k) (1 - \rho_c)^k \rho_c^{\mathcal{E}-k}, \quad (51)$$

where $\mathcal{E}$ is the number of communication edges, $F(k)$ denotes the number of connected subgraphs obtained when exactly k edges are preserved, and ρ_c is the edge failure probability. The notation ρ_c is used here to avoid conflict with the finite-time exponent p .

Equation (51) should be interpreted carefully. If T_s is an expected value, it cannot by itself prove deterministic finite-time convergence. A deterministic theorem requires a lower bound on the accumulated trusted connectivity in (40). In contrast, a probabilistic theorem can be stated in terms of

$$\mathbb{P} \left\{ \int_0^{t_f} \lambda_2(L_r(s)) ds \geq \frac{1}{2k} \ln \frac{V(0)}{\varepsilon} \right\}. \quad (52)$$

This probability quantifies the likelihood that the vehicles complete time-to-go synchronization before interception under random link failures.

X. SIMULATION PLAN

The following cases can be used to verify the proposed framework.

A. No-Attack Baseline

The first case verifies that the proposed guidance law reduces to the nominal cooperative guidance law when $\gamma_{ij} = 1$ and $f_{ij} = 0$ for all links.

B. DoS-Only Case

Several communication links are blocked over prescribed time intervals. The zero-bias strategy, the holding-type virtual leader, and the target-oriented virtual leader should be compared.

C. FDI-Only Case

Bias, sinusoidal, state-dependent, and slow-drift FDI signals are injected into selected links. The physical trust $\rho_{ij}^{\text{phy}}(t)$, the group-consistency or GMM trust $\rho_{ij}^{\text{gmm}}(t)$, the distribution trust $\rho_{ij}^{\text{kl}}(t)$, and the final binary isolation decision $\sigma_{ij}(t)$ should be plotted to show when corrupted links are removed from the trusted graph. This case should also compare the proposed multi-evidence isolation rule with a single residual detector.

D. Hybrid DoS and FDI Case

Some links are blocked by DoS, while other available links are corrupted by FDI. This case is the main scenario for validating the trusted graph construction in (23), where DoS-blocked and FDI-detected links are both excluded before the cooperative guidance command is computed.

E. Boundary Test on Effective Communication Time

The DoS duty cycle and FDI intensity are varied to test whether the convergence behavior changes consistently with the cumulative trusted-connectivity conditions (40) and (48).

F. Ablation Study

The ablation study should remove one module at a time: physical feasibility, group-consistency/GMM trust, windowed residual statistics, link isolation, node isolation, and zero-bias fallback. The expected metrics include miss distance, impact-time disagreement, $V(t)$, $\int_0^t \lambda_2(L_r(s)) ds$, trust-weight evolution, false alarm rate, missed detection rate, isolated-link count, and command magnitude.

XI. POSSIBLE CONTRIBUTIONS

The method can be written as the following contributions:

- 1) A hybrid DoS-FDI attack model is established for cooperative guidance, simultaneously describing missing neighbor information and corrupted neighbor information.

- 2) A multi-evidence FDI trust evaluation mechanism is proposed by combining physical feasibility, group-consistency clustering, and windowed distribution divergence, reducing the dependence on a single prediction residual.
- 3) A trusted-graph isolation mechanism is proposed: DoS-blocked links and FDI-detected links are removed from the cooperative graph before they can affect the guidance command.
- 4) Exponential and fixed-time convergence conditions are derived in terms of accumulated trusted connectivity, clarifying when synchronization before interception can still be guaranteed under hybrid cyber attacks.

XII. TECHNICAL BOUNDARIES

Several limitations should be stated explicitly in the final manuscript. First, the physical feasibility interval in (9) must be constructed from valid bounds on acceleration, target maneuver, and field-of-view constraints; otherwise the physical trust score may become overly conservative. Second, GMM-based or median-based trust requires enough neighbor redundancy to separate normal and suspicious links. If too many vehicles are attacked simultaneously, the isolation result may remove too much of the communication graph. Third, windowed residual statistics depend on the quality of the nominal

residual model and the window length. A short window reacts quickly but may increase false alarms, whereas a long window is smoother but slower. Fourth, link isolation protects the controller from detected corrupted data, but convergence still requires sufficient trusted connectivity after isolation. Finally, if field-of-view or nonsingularity auxiliary functions drive the effective impact-time control gain to zero, fixed-time convergence cannot be strictly guaranteed. In that case, the result should be weakened to asymptotic convergence, convergence to a small neighborhood, or finite-time convergence under an additional gain lower-bound condition.

XIII. CONCLUSION

This draft provides a complete framework for handling DoS and FDI attacks in cooperative guidance. The key is to avoid allowing unavailable or corrupted neighbor data to enter the cooperative guidance law. DoS is handled as link unavailability, whereas detected FDI is handled by isolating the corrupted link or, in conservative cases, the compromised node. The resulting trusted graph is used to construct the impact-time synchronization error and the cooperative guidance command. Under sufficient accumulated trusted connectivity after isolation, the predicted impact times converge before interception or reach a prescribed disagreement bound at the terminal time.